

MicroCluster

A privacy-preserving community health-signal system, with a simulator that measures what the privacy costs.

It detects the signal without identifying the reporter, and won't name a group too small or too covered to name safely.

THE PROBLEM

A cluster of the same illness on a dorm or office floor is worth catching early, but surfacing it means someone reports being sick and says where they are. The useful signal and the privacy risk are the same data.

WHAT IT DOES

People submit anonymous reports from a fixed checklist: one of four symptom categories, a coarse onset, one location group. No name, no free text, no room. Two detectors watch the stream, one relative to a location's neighbours, one against a background rate; when one fires, a disclosure engine decides what may be said and at what scope. A headless simulator generates the reports and measures the policy.

THE MECHANISM: TWO GATES, PULLING OPPOSITE WAYS

A scope is named only if it clears both gates, evaluated separately, never merged into one score:

- **Statistical gate:** qualifying reports $\geq \max(5, \lceil \sqrt{n} \rceil)$ for declared population n . Scales *up* with n : a bigger claim needs more evidence.
- **Privacy gate:** $n \geq 20$, and qualifying / $n \leq 0.5$. Scales *down* with n : a small, near-fully-covered group is the identifying one, and naming it publishes a roster.

The system discloses the finest scope clearing both; if none does, it says nothing. Report counts render as bands ("5–9", "10–19"), never exact integers.

RESULTS — 500-SEED SWEEP, 300-SEED BENCHMARK, 200-SEED SELF-TEST

The delay the policy buys (detection fixed, disclosure gate tightened; the detector fires around day 8 in both rows).

Disclosure gate	First disclosure	Infected by then	Finest scope
shipped	day 14	~35	one suite
tightest tested	day 19	~82	floor only

Authored rules vs a learned model, held out by run.

Detector	Precision	Recall	Mean delay
authored threshold rules	0.95	0.78	3.7 days
logistic-regression model	0.93	0.82	2.8 days

The detector fires around day 8; disclosure is not permitted until about day 14, and that six-day gap, roughly 35 people infected, is the measurable cost of the policy. The learned model is about a day faster for about three points of precision, so the authored rules ship and the model stays a benchmark. Differencing the banded output pins an exact one-person overnight change on 0 days, against 358 for exact counts; the direction of change still leaks on 204 of 1,171 same-scope day-pairs, reported not hidden.

LIMITATIONS — READ THIS

Every transmission, reporting, contact and incubation value is chosen for demonstration, not measured; there is no real dataset, and the system is validated only at one population size (150). The detector is exercised only against a clean, memoryless reporting model, and fires on roughly 60% of outbreaks that never establish. It predicts nothing about a real building and treats no one; the claim is understanding, not intervention. It rate-limits submissions per session but does not authenticate reporters, so a determined attacker using multiple sessions can still inject signal. Agent-based epidemic modelling and small-cell suppression are both mature prior art. The contribution is wiring the suppression policy directly onto a live anomaly detector as one server-side pipeline, and measuring the trade-off it forces.

Tech. Python, FastAPI + Uvicorn; detection and disclosure run server-side only. Plain HTML/CSS/JS, hand-drawn SVG chart, no framework or build step. scikit-learn for the offline benchmark only. 223 tests green; contrast-checked, keyboard-navigable.

Links. Repository github.com/koridoesstuff/MicroCluster · Live demo microcluster.onrender.com · Video

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